

IF330 – Pemrograman Web

12 - Forms & Validation

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PROGRAM STUDI INFORMATIKA
FAKULTAS TEKNIK DAN INFORMATIKA
UNIVERSITAS MULTIMEDIA NUSANTARA
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REVIEW



- Eloquent Relationship
 - 1-to-1
 - 1-to-many
 - many-to-many

TODAY's OUTLINE

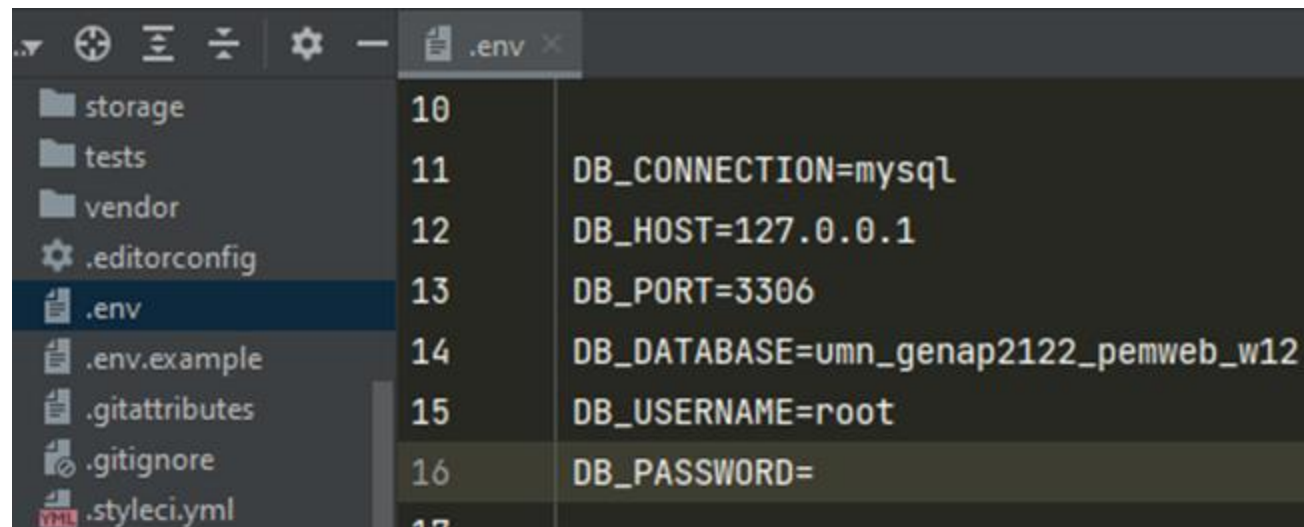
- Forms
- Validation
- File Upload

A Simple CRUD App

What you should prepare:

- Database Setup on .env file
- Create Model & Migration file
 - Setup the database table
 - Migrate
- Create Controller
- Setup Route
- Create Views

Database Setup on .env file



The screenshot shows a code editor with a sidebar on the left containing a file explorer. The files listed are storage, tests, vendor, .editorconfig, .env (selected), .env.example, .gitattributes, .gitignore, and .styleci.yml. The main editor area displays the contents of the .env file, which includes database connection settings for MySQL.

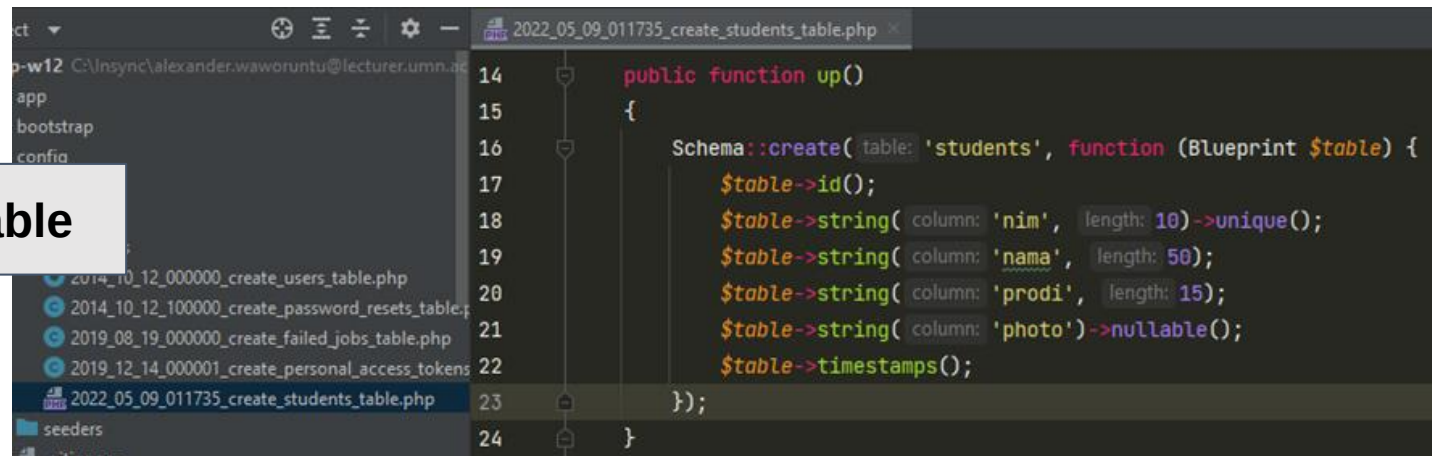
```
10  
11 DB_CONNECTION=mysql  
12 DB_HOST=127.0.0.1  
13 DB_PORT=3306  
14 DB_DATABASE=umn_genap2122_pemweb_w12  
15 DB_USERNAME=root  
16 DB_PASSWORD=  
17
```

Create Model & Migration File

Create Model & Migration with php artisan

```
2021-22\Pemrograman-Web\Week-12\app-w12> php artisan make:model Student -m  
Model created successfully.  
Created Migration: 2022_05_09_011735_create_students_table
```

Setup the database table



```
14 public function up()  
15 {  
16     Schema::create( table: 'students', function (Blueprint $table) {  
17         $table->id();  
18         $table->string( column: 'nim', length: 10)->unique();  
19         $table->string( column: 'nama', length: 50);  
20         $table->string( column: 'prodi', length: 15);  
21         $table->string( column: 'photo')->nullable();  
22         $table->timestamps();  
23     });  
24 }
```

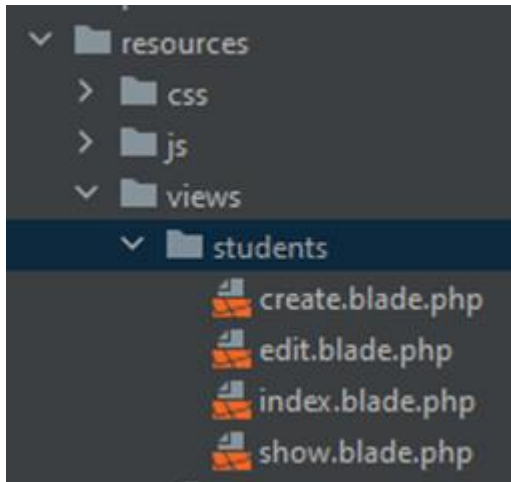
Migrate

```
2021-22\Pemrograman-Web\Week-12\app-w12> php artisan migrate  
Migration table created successfully.  
Migrating: 2019_08_19_000000_create_failed_jobs_table  
Migrated: 2019_08_19_000000_create_failed_jobs_table (60.21ms)  
Migrating: 2019_12_14_000001_create_personal_access_tokens_table  
Migrated: 2019_12_14_000001_create_personal_access_tokens_table (77.41ms)  
Migrating: 2022_05_09_011735_create_students_table  
Migrated: 2022_05_09_011735_create_students_table (61.87ms)
```

Create Controller, Route & View

```
2021-22\Pemrograman-Web\Week-12\app-w12> php artisan make:controller StudentController --resource
```

```
Route::resource(name: 'students', controller: StudentController::class);
```

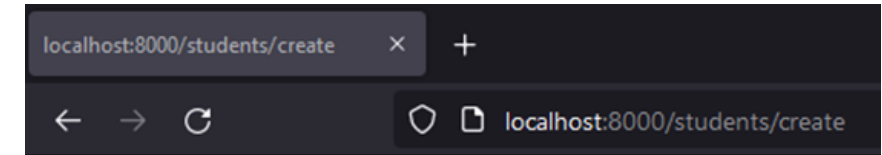


```
rograman-Web\Week-12\app-w12> php artisan route:list
```

```
GET|HEAD / .....  
POST _ignition/execute-solution ignition.executeSolution > Spatie\LaravelIg...  
GET|HEAD _ignition/health-check ignition.healthCheck > Spatie\LaravelIgnition ...  
POST _ignition/update-config ignition.updateConfig > Spatie\LaravelIgnition...  
GET|HEAD api/user .....  
GET|HEAD sanctum/csrf-cookie ..... Laravel\Sanctum > CsrfCookieController@show  
GET|HEAD students ..... students.index > StudentController@index  
POST students ..... students.store > StudentController@store  
GET|HEAD students/create ..... students.create > StudentController@create  
GET|HEAD students/{student} ..... students.show > StudentController@show  
PUT|PATCH students/{student} ..... students.update > StudentController@update  
DELETE students/{student} ..... students.destroy > StudentController@destroy  
GET|HEAD students/{student}/edit ..... students.edit > StudentController@edit
```


CREATE Data

```
Project create.blade.php
1 <h1>Create New Student</h1>
2 <form action="/students" method="post">
3     @csrf
4     NIM: <input type="text" name="nim" /><br />
5     Nama: <input type="text" name="nama" /><br />
6     Prodi: <input type="text" name="prodi" /><br />
7     <button type="submit">Submit</button>
8 </form>
```



Create New Student

NIM:

Nama:

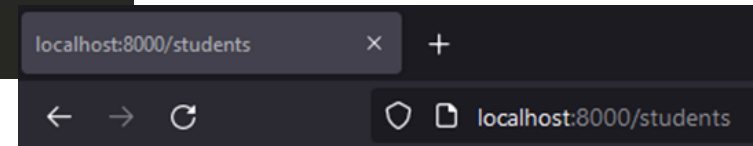
Prodi:

```
StudentController.php
21
22 public function store(Request $request)
23 {
24     $student = new Student();
25     $student->nim = $request->nim;
26     $student->nama = $request->nama;
27     $student->prodi = $request->prodi;
28     $student->save();
29     return "Berhasil menyimpan data Student dengan id=" . $student->id;
30 }
```


RETRIEVE All Data

```
StudentController.php
10
11 public function index()
12 {
13     $students = Student::all();
14     return view('students.index', ['students' => $students]);
15 }
16
```

```
index.blade.php
9 </tr>
10 @foreach($students as $student)
11 <tr>
12 <td>{{ $student->nim }}</td>
13 <td>{{ $student->nama }}</td>
14 <td>{{ $student->prodi }}</td>
15 <td>
16 <a href="/students/{{ $student->id }}">SHOW</a> |
17 <a href="/students/{{ $student->id }}/edit">EDIT</a> |
18 DELETE
19 </td>
20 </tr>
21 @endforeach
22 </table>
```



Daftar Mahasiswa

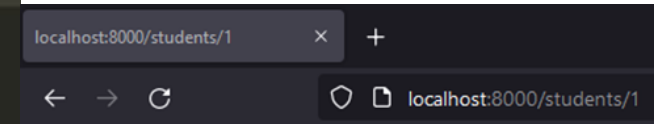
[Create new Student](#)

NIM	Nama	Prodi	Tindakan
001	John Thor	IF	SHOW EDIT DELETE
002	John Wick	TK	SHOW EDIT DELETE

RETRIEVE Single Data

```

StudentController.php
34 public function show($id)
35 {
36     $student = Student::findOrFail($id);
37     return view('students.show', ['student' => $student]);
38 }
39
40 public function edit($id)
  
```



John Thor

```

show.blade.php
1 <h1>{{$student->nama}}</h1>
2 <a href="/students">< Back to Student List</a>
3 <br />
4 NIM: {{$student->nim}}<br />
5 Nama: {{$student->nama}}<br />
6 Prodi: {{$student->prodi}}<br />
7 Photo: <img src="" />
  
```

[< Back to Student List](#)
 NIM: 001
 Nama: John Thor
 Prodi: IF
 Photo:

EDIT Data

```

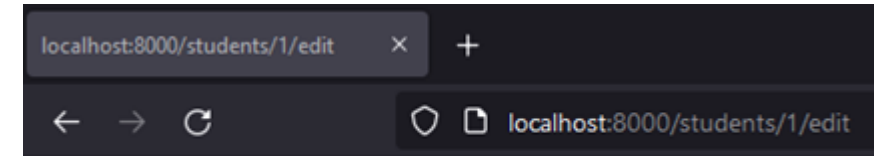
StudentController.php
39 public function edit($id)
40 {
41     $student = Student::findOrFail($id);
42     return view('students.edit', ['student' => $student]);
43 }
  
```

```

edit.blade.php
1 <h1>Edit Student</h1>
2 <form action="/students/{{ $student->id }}" method="post">
3     @method('PUT')
4     @csrf
5     NIM: <input type="text" name="nim" value="{{ $student->nim }}" /><br />
6     Nama: <input type="text" name="nama" value="{{ $student->nama }}" /><br />
7     Prodi: <input type="text" name="prodi" value="{{ $student->prodi }}" /><br />
8     <button type="submit">Submit</button>
9 </form>
  
```

```

StudentController.php
45 public function update(Request $request, $id)
46 {
47     $student = Student::findOrFail($id);
48     $student->nim = $request->nim;
49     $student->nama = $request->nama;
50     $student->prodi = $request->prodi;
51     $student->save();
52     return redirect(to: '/students');
53 }
  
```



Edit Student

NIM:

Nama:

Prodi:

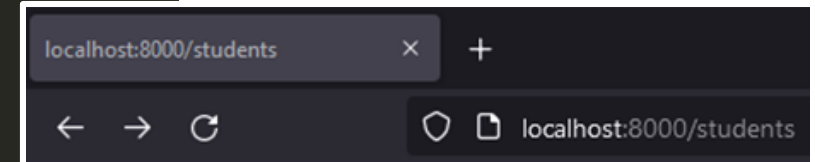
DELETE Data

```

StudentController.php x index.blade.php x
13 <td>{{$student->nama}}</td>
14 <td>{{$student->prodi}}</td>
15 <td>
16 <a href="/students/{{$student->id}}">SHOW</a> |
17 <a href="/students/{{$student->id}}/edit">EDIT</a> |
18 <form action="/students/{{$student->id}}" method="post">
19 @method('DELETE')
20 @csrf
21 <button type="submit">DELETE</button>
22 </form>
23 </td>
  
```

```

StudentController.php x
54
55 public function destroy($id)
56 {
57     $student = Student::find($id);
58     $student->delete();
59     return redirect( to: '/students');
60 }
61
  
```



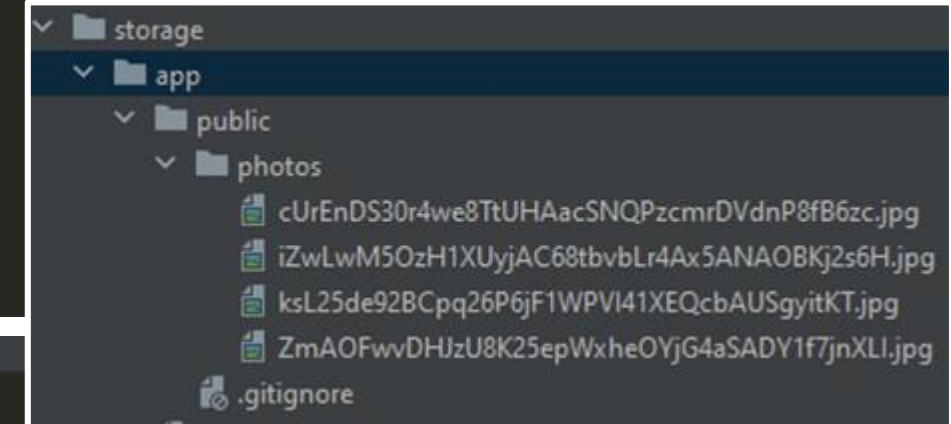
Daftar Mahasiswa

[Create new Student](#)

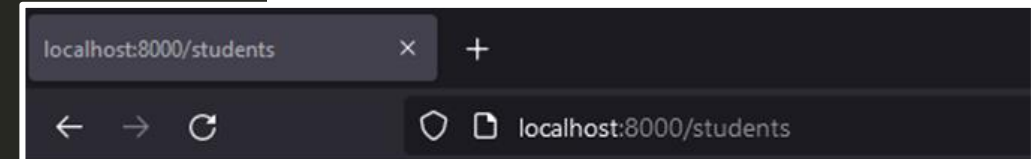
NIM	Nama	Prodi	Tindakan
002	John Wick	TK	SHOW EDIT DELETE

File Upload

```
create.blade.php
1 <h1>Create New Student</h1>
2 <form action="/students" method="post" enctype="multipart/form-data">
3     @csrf
4     NIM: <input type="text" name="nim" /><br />
5     Nama: <input type="text" name="nama" /><br />
6     Prodi: <input type="text" name="prodi" /><br />
7     Foto: <input type="file" name="photo" />
8     <button type="submit">Submit</button>
9 </form>
```

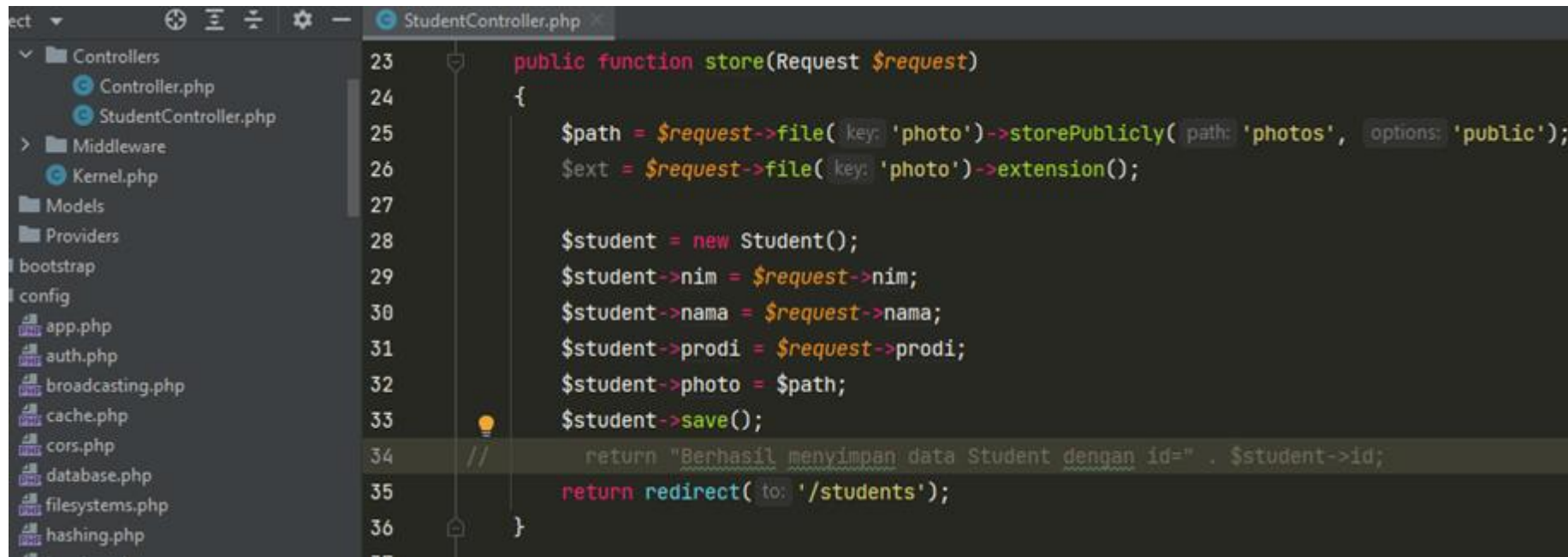


```
StudentController.php
23 public function store(Request $request)
24 {
25     $path = $request->file( key: 'photo' )->storePublicly( path: 'photos', options: 'public' );
26     $ext = $request->file( key: 'photo' )->extension();
27
28     return "File stored at: " . $path . "<br />File Extension: " . $ext;
29
30     // $student = new Student();
31     // $student->nim = $request->nim;
32     // $student->nama = $request->nama;
33     // $student->prodi = $request->prodi;
34     // $student->photo = $path;
35     // $student->save();
36     // return "Berhasil menyimpan data Student dengan id=" . $student->id;
37     // return redirect('/students');
```



File stored at: photos/iZyXRVOHRphc0iSMFoxXJRSttwTazU889ydFMuPQ.jpg
File Extension: jpg

Save Uploaded File Path to Database

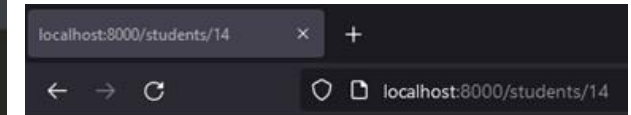


```
23 public function store(Request $request)
24 {
25     $path = $request->file( key: 'photo')->storePublicly( path: 'photos', options: 'public');
26     $ext = $request->file( key: 'photo')->extension();
27
28     $student = new Student();
29     $student->nim = $request->nim;
30     $student->nama = $request->nama;
31     $student->prodi = $request->prodi;
32     $student->photo = $path;
33     $student->save();
34     // return "Berhasil menyimpan data Student dengan id=" . $student->id;
35     return redirect( to: '/students');
36 }
```

Displaying File URL to View

```
StudentController.php
38 public function show($id)
39 {
40     $student = Student::findOrFail($id);
41     $photo = Storage::url($student->photo);
42     return view('students.show', ['student' => $student, 'photo' => $photo]);
43 }
```

```
show.blade.php
1 <h1>{{ $student->nama }}</h1>
2 <a href="/students">< Back to Student List</a>
3 <br />
4 NIM: {{ $student->nim }}<br />
5 Nama: {{ $student->nama }}<br />
6 Prodi: {{ $student->prodi }}<br />
7 Photo: 
8
```



John Thor

[< Back to Student List](#)

NIM: 001

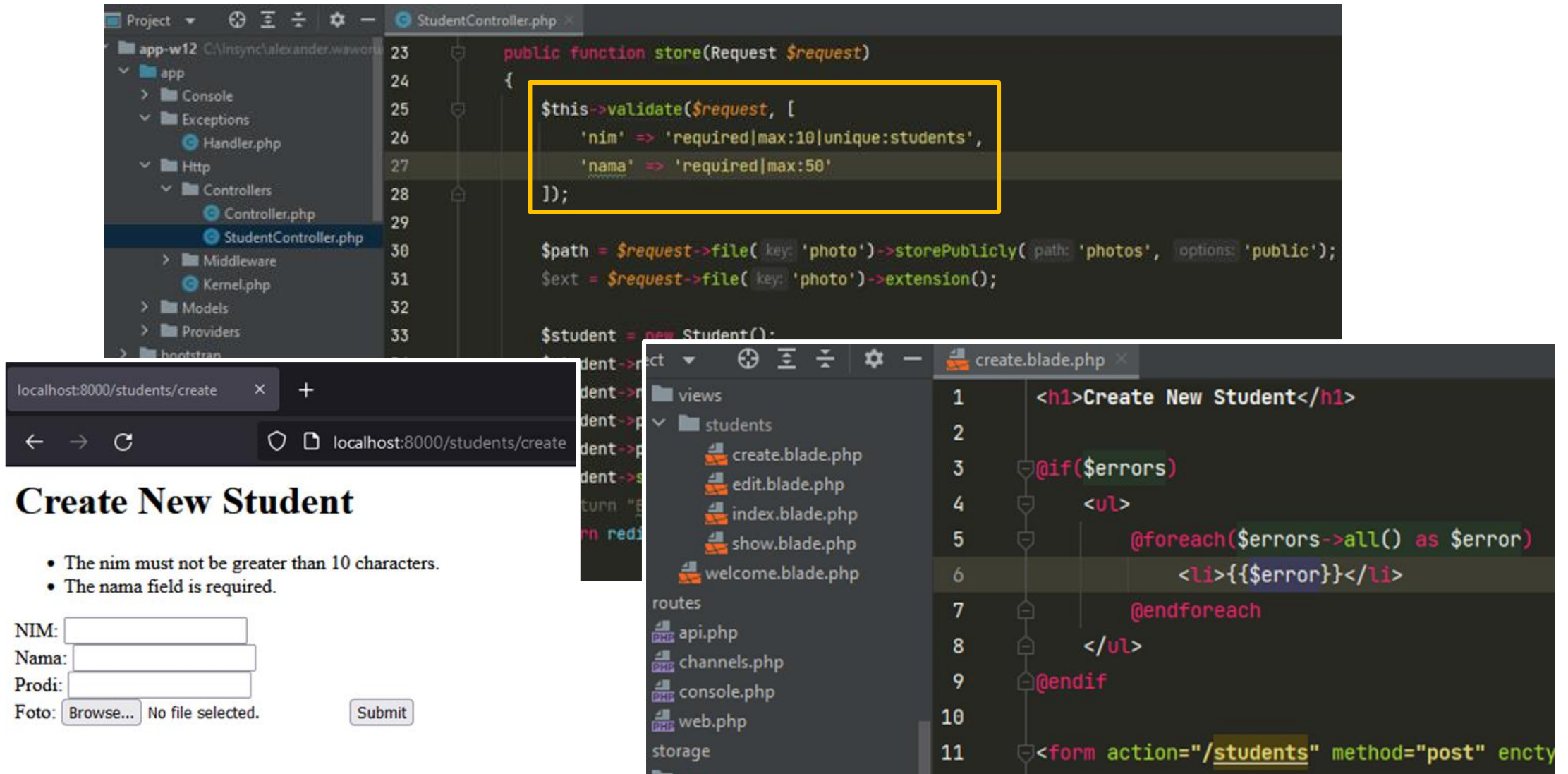
Nama: John Thor

Prodi: IF



Photo:

Form Validation



The image displays a web application interface for creating a new student, along with the underlying PHP code and Blade template.

Browser Window: The address bar shows `localhost:8000/students/create`. The page title is "Create New Student".

Form Fields:

- NIM:
- Nama:
- Prodi:
- Foto: No file selected.

Validation Rules (from `StudentController.php`):

```
public function store(Request $request)
{
    $this->validate($request, [
        'nim' => 'required|max:10|unique:students',
        'nama' => 'required|max:50'
    ]);

    $path = $request->file('photo')->storePublicly(path: 'photos', options: 'public');
    $ext = $request->file('photo')->extension();

    $student = new Student();
    $student->nim = $request->input('nim');
    $student->nama = $request->input('nama');
    $student->prodi = $request->input('prodi');
    $student->foto = $path . $ext;
    $student->save();

    return redirect('/students');
}
```

Blade Template (from `create.blade.php`):

```
<h1>Create New Student</h1>
<div>
    @if($errors)
        <ul>
            @foreach($errors->all() as $error)
                <li>{{ $error }}</li>
            @endforeach
        </ul>
    @endif
</div>
<form action="/students" method="post" enctype="multipart/form-data">
    <input type="text" value="{{ $nim }}" />
    <input type="text" value="{{ $nama }}" />
    <input type="text" value="{{ $prodi }}" />
    <input type="button" value="Browse..." />
    <input type="button" value="Submit" />
</form>
```

SUMMARY

After finishing this module, you should be able:

- to understand how to implement forms with Laravel
- to understand how to create CRUD App with Laravel
- to understand how to implement file upload
- to understand how to implement form validation

NEXT WEEK'S OUTLINE



- Laravel Security
 - Middleware
 - User Authentication

REFERENCES

<https://laravel.com/docs/9.x>

Visi

Menjadi Program Studi Strata Satu Informatika **unggulan** yang menghasilkan lulusan **berwawasan internasional** yang **kompeten** di bidang Ilmu Komputer (*Computer Science*), **berjiwa wirausaha** dan **berbudi pekerti luhur**.



Misi

1. Menyelenggarakan pembelajaran dengan teknologi dan kurikulum terbaik serta didukung tenaga pengajar profesional.
2. Melaksanakan kegiatan penelitian di bidang Informatika untuk memajukan ilmu dan teknologi Informatika.
3. Melaksanakan kegiatan pengabdian kepada masyarakat berbasis ilmu dan teknologi Informatika dalam rangka mengamalkan ilmu dan teknologi Informatika.